

## Overview

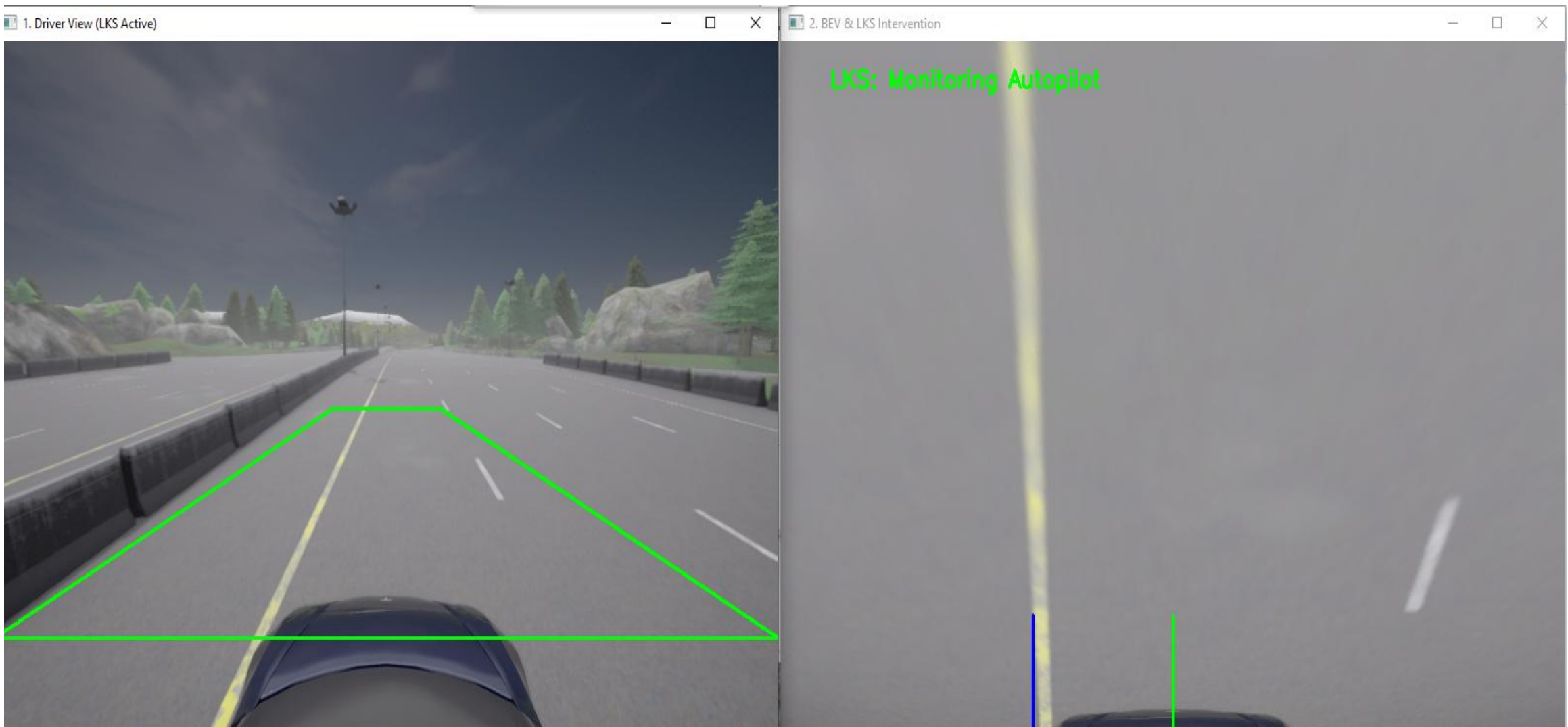
This project develops a vision-based Lane Keeping Support (LKS) system in the CARLA simulator to detect boundaries and prevent lane departures. Using Inverse Perspective Mapping (IPM), it transforms front-camera pixels into a bird's-eye view for precise vehicle control. The system calculates real-time lateral deviation to provide minimal steering interventions, dynamically applying braking and counter-steering to stabilize the vehicle before returning control to the autopilot.

## System Overview

The system utilizes a forward-facing RGB camera mounted on an autonomous vehicle within the CARLA simulator. To correct the camera's natural perspective distortion, the system applies an Inverse Perspective Mapping (IPM) algorithm, transforming a predefined Region of Interest (ROI) into a flat, 2D Bird's Eye View. By applying binary thresholding and calculating image moments on this flattened view, the system accurately extracts the lane markings and calculates the vehicle's lateral deviation from the lane's true center.

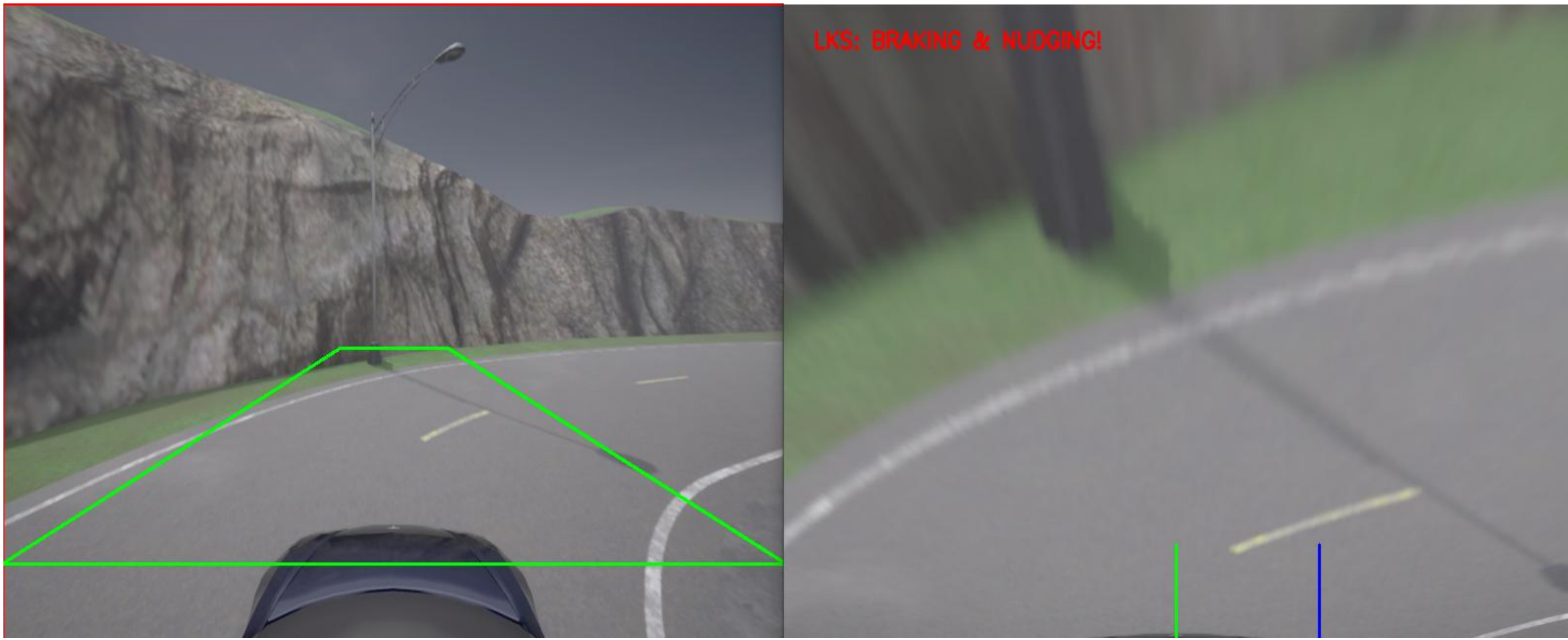
## RT Monitoring & Visualization

A custom split-screen OpenCV visualization module was developed to monitor the system's perception and decision-making in real-time. The interface displays the original driver's view overlaid with the dynamic green ROI trapezoid. Adjacent to it, the Bird's Eye View visualizes the computed lane center, the vehicle's trajectory, and live textual alerts detailing the active LKS state (Monitoring, Nudging, or Straightening), allowing for rapid debugging and parameter tuning.



## Project Impact

This project successfully demonstrates the integration of geometric computer vision algorithms with active vehicle control in a highly realistic simulation environment. The system achieves robust lane keeping through minimal, targeted interventions, highlighting a highly practical approach to Advanced Driver Assistance Systems (ADAS) that prioritizes both passenger safety and ride smoothness.



## Control Mechanism

The LKS logic uses a Proportional (P) controller and a state machine for subtle interventions across three states:

- **Monitoring:** Remains passive, allowing the default autopilot to navigate safely.
- **Braking & Nudging:** On critical lane departure, it cuts throttle, applies 40% braking, and steers proportionally.
- **Straightening:** Applies a brief counter-steer to prevent overcorrection before seamlessly returning control.

[Link to the Carla Demo](#)



[Link to the LKS Demo](#)

